

The Classical Hall Effect

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Abstract

This experiment investigated the Hall effect in semiconductor materials. The primary objectives were to determine the Hall coefficient (R_H), charge carrier concentration (n), and carrier mobility (μ) for n-type and p-type Bismuth (Bi) samples at room temperature, and to study the temperature dependence of the Hall coefficient for the p-type sample. A known current was passed through the samples while they were subjected to a perpendicular magnetic field, and the resulting transverse Hall voltage (V_H) was measured. At room temperature, the Hall coefficients for the n-type and p-type samples were found to have opposite signs, confirming their respective majority charge carriers. In the temperature-dependent study of p-type Bi, the Hall coefficient was observed to be positive at low temperatures, then decrease, pass through zero, and become negative as temperature increased, demonstrating the phenomenon of sign inversion. This inversion is attributed to the thermal generation of intrinsic electron-hole pairs and the higher mobility of electrons, which dominate the conduction process at elevated temperatures.

1 Introduction

The Hall effect is a fundamental phenomenon in solid-state physics that provides critical insights into the electronic properties of conducting materials. Discovered by Edwin Hall in 1879, the effect describes the generation of a transverse voltage (the Hall voltage) across a current-carrying conductor when it is placed in a perpendicular magnetic field. While standard resistivity measurements can determine a material's opposition to current flow, they cannot by themselves reveal the type (positive holes or negative electrons) or concentration of the charge carriers responsible for conduction. The Hall effect experiment is a powerful tool that allows for the direct determination of both of these crucial parameters.

When a current (I_x) flows through a semiconductor sample of thickness t placed in a magnetic field (H) perpendicular to the current, the charge carriers experience a Lorentz force. This force pushes the carriers—electrons or holes—to one side of the sample, creating an accumulation of charge. This charge separation establishes a transverse electric field, the Hall field (E_H), which opposes the further migration of carriers. At equilibrium, the electric force from this field balances the magnetic Lorentz force, resulting in a stable, measurable potential difference known as the Hall voltage (V_H).

This Hall voltage is related to the Hall coefficient (R_H) of the material by the equation:

$$R_H = \frac{V_H \cdot t}{I_x \cdot H}$$

Theoretically, for a simple model with one type of charge carrier, the Hall coefficient is inversely proportional to the carrier concentration (n) and the elementary charge (e):

$$R_H = \frac{1}{n \cdot e}$$

Crucially, the sign of the charge e determines the sign of the Hall coefficient. For n-type semiconductors, where electrons are the majority carriers (e is negative), R_H is negative. For p-type semiconductors, where holes are the majority carriers (e is positive), R_H is positive. Thus, a simple measurement of the Hall voltage sign immediately identifies the material type.

Furthermore, the Hall coefficient, in combination with the material's conductivity (σ), can be used to calculate the carrier mobility (μ), which describes how easily charge carriers move through the material:

$$\mu = |R_H| \cdot \sigma$$

The properties of semiconductors are also strongly dependent on temperature. In a p-type semiconductor, increasing temperature can thermally excite electrons from the valence band to the conduction band, creating intrinsic electron-hole pairs. Although the material is doped p-type, the mobility of electrons is generally much higher than that of holes. At a high enough temperature (the inversion temperature), these thermally generated, high-mobility electrons can come to dominate the conduction process, causing the Hall coefficient to decrease, pass through zero, and eventually become negative.

This experiment has two primary objectives:

1. To measure the Hall coefficient for n-type and p-type germanium samples at room temperature, and from this data, determine the majority carrier type, concentration, and mobility for each.
2. To measure the Hall coefficient for a p-type germanium sample as a function of increasing temperature, thereby observing and quantifying the phenomenon of sign inversion.

2 Theory

Conductivity measurements in semiconductors cannot reveal whether one or both types of carriers are present, nor distinguish between them. However, this information can be obtained from Hall Effect measurements, which are a basic tool for the determination of mobilities. The effect was discovered by E.H. Hall in 1879.

Consider a simple crystal mounted as in the Fig. 4, with a magnetic field H in the z direction perpendicular to contacts 1, 2 and 3, 4. If current is flowing through the crystal in the x direction (by application of a voltage V_x between contacts 1 and 2), a voltage will appear across contacts 3, 4 in the y - direction. It is easy to calculate this (Hall) voltage if it is assumed that all carriers have the same drift velocity. We will do this in two steps: (a) by assuming that carriers of only one type are present, and (b) by assuming that carriers of both types are present.

(a) One type of carrier

The magnetic force on the carriers is $\vec{F}_m = e\vec{E}_m = e(\vec{v} \times \vec{H})$ and it is compensated by the force $\vec{F}_H$ due to the Hall field $\vec{E}_H$, $\vec{F}_H = e\vec{E}_H = \vec{F}_m$. As $\vec{v}$ is along the x - axis and $\vec{H}$ along the z -axis, the electric field $\vec{E}_m$ is along the y -axis and is given by $E_m = vH = \mu E_x H$ where μ is the carrier mobility given by $v = \mu E_x$ and E_x is the applied electric field along the x -axis. This electric field is related to the current density and conductivity, $E_x = J_x/\sigma$. The Hall coefficient R_H is defined as

$$|R_H| = \frac{E_m}{J_x H} = \frac{\mu E_x}{J_x} = \frac{\mu}{\sigma} = \frac{1}{ne}$$

We have used the relation $\sigma = ne\mu$. Hence for fixed magnetic field and fixed input current, the Hall voltage is proportional to $1/n$. It follows that

$$\mu = R_H \sigma,$$

providing an experimental measurement of the mobility. R_H is expressed in $\text{cm}^3 \text{ coulomb}^{-1}$.

Experimentally the coefficient is given by

$$R_H = \frac{V_y b}{(I_x / bt) H} = \frac{V_y t}{I_x H}$$

where b and t are respectively the width and the thickness of the sample. In case the voltage across the input is kept constant, it is convenient to define the Hall angle as the ratio of applied and measured voltages:

$$\phi = \frac{V_y}{V_x} = \frac{E_y b}{E_x l} = \mu \frac{b}{l} H$$

where ' l ' is the length of the crystal. The Hall angle is thus proportional to the mobility.

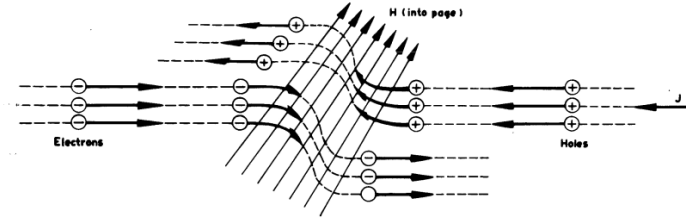


Figure 1: Carrier separation due to a magnetic field

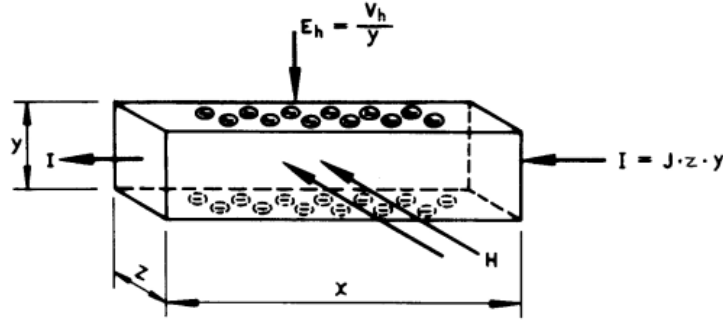


Figure 2: Sample for studying Hall Effect

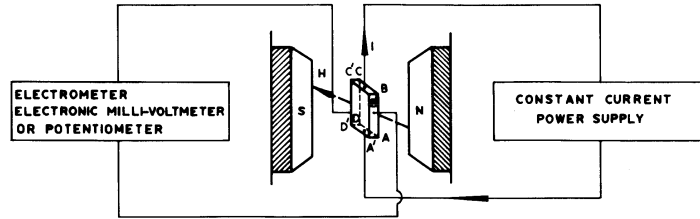


Figure 3: Schematic of experimental setup (Electromagnet, Constant Current Power Supply, and Voltmeter).

(b) Two types of carriers

Now it is important to recognize that for the same electric field E_x , the Hall voltage for p carriers (holes) will have opposite sign from that for n carriers (electrons). (That is, the Hall coefficient R has a different sign.) Thus, the Hall field E_y will not be able to compensate for

the magnetic force on both types of carriers and there will be a transverse motion of carriers; however, the net transverse transfer of charge will remain zero since there is no current through the 3, 4 contacts; this statement is expressed as

$$e(v_y^+ p - v_y^- n) = 0$$

while

$$e(v_x^+ p - v_x^- n) = J_x \quad \text{and} \quad e(\mu^+ p + \mu^- n) = \sigma$$

where the mobility is always a positive number; however, v_x^+ has the opposite sign from v_x^- . It is given by

$$v = \frac{s}{\tau} = \left(\frac{1}{2} \frac{F}{m^*} \tau^2 \right) \frac{1}{\tau}$$

where τ is mean time between collisions and m^* is the effective mass of the carriers. Now for holes and electrons, we have

$$\vec{F}^+ = e[(\vec{v}_x^+ \times \vec{H}) - \vec{E}_H]$$

$$\vec{F}^- = -e[(\vec{v}_x^- \times \vec{H}) - \vec{E}_H]$$

If m_h and m_e are effective masses for holes and electrons, respectively, we get

$$v_y^+ = \frac{1}{2} \frac{e}{m_h} \tau [(\mu^+ E_x H) - E_H] = \mu^+ (\mu^+ E_x H - E_H)$$

$$v_y^- = \frac{1}{2} \frac{e}{m_e} \tau [(\mu^- E_x H) - E_H] = \mu^- (\mu^- E_x H + E_H)$$

$$\mu^+ p (\mu^+ E_x H - E_H) - \mu^- n (\mu^- E_x H + E_H) = 0$$

or

$$E_H = E_x H \frac{(\mu_h^2 p - \mu_e^2 n)}{\mu_h p + \mu_e n}$$

and for the Hall coefficient R_H

$$R_H = \frac{E_H}{J_x H} = \frac{E_H}{\sigma E_x H} = \frac{\mu_h^2 p - \mu_e^2 n}{e(\mu_h p + \mu_e n)^2}$$

Equation (6) correctly reduces to Eq. (1) when only one type of carrier is present.

Since the mobilities μ_h and μ_e are not constants but functions of T, the Hall coefficient given by Eq. (6) is also a function of T and it may become zero and even change sign. In general $\mu_e > \mu_h$ so that inversion may happen only if $p > n$; thus "Hall coefficient inversion" is characteristic of only "p-type" semiconductors.

At the point of zero Hall Coefficient, it is possible to determine the ratio of mobilities μ_e/μ_h in a simple manner.

3 Apparatus

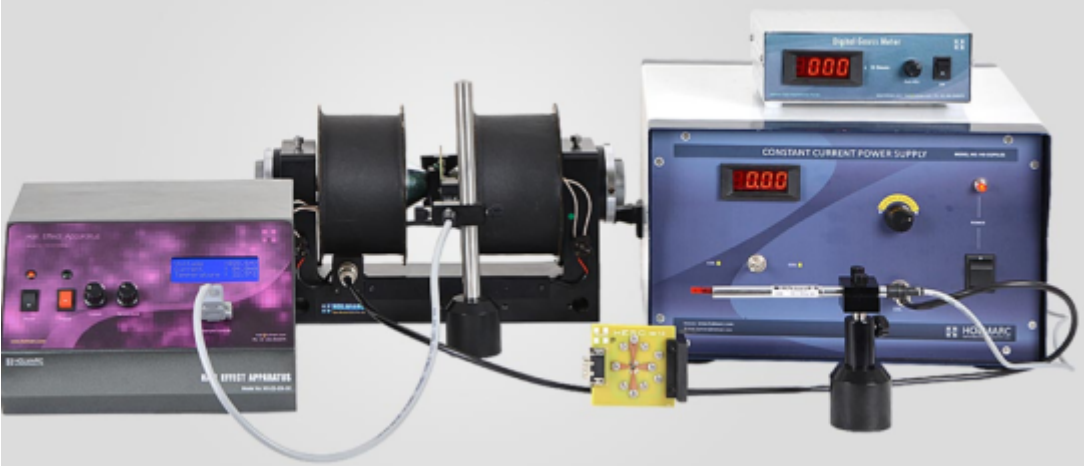


Figure 4: Schematic arrangement for the measurement of Hall Effect of a crystal

1. Hall probe (Ge: p-type, Ge: n-type, Si: n-type): Ge/Si single crystal with four spring type pressure contact is mounted on a glass-epoxy strips. Four leads are provided for connections with the probe current and Hall voltage measuring devices.
2. Oven
3. Temperature sensor: Temperature is measured with Cromel-Alumel thermocouple with its junction at a distance of 1 mm from the crystal.
4. Hall Effect Set-up, Model: DHE-22 The set-up, DHE-22 consists of two sub set-ups, each consisting of further two units.
 - (a) Measurement of Probe Current & Hall Voltage: This unit consists of a digital millivoltmeter and constant current power supply. The Hall voltage and probe current can be read on the same digital panel meter through a selector switch.
 - (b) Measurement of Thermo emf and Heater current: The unit consists of a digital millivoltmeter and constant current power supply. The thermo emf of thermocouple and heater current can be read on the same DPM through a selector switch.

4 Procedure

The Gaussmeter probe was placed in the electromagnet's gap. The electromagnet current (I) was increased in steps, and the corresponding magnetic field (H_z) was recorded. A calibration graph of H_z versus I was plotted to determine the field strength during the experiment. Step 2 was repeated in its entirety using the p-type Ge sample to determine its R_H , carrier concentration (p), and mobility (μ_h).

5 Data

Table 1: Combined Raw and Derived Hall Effect Data

(a) Raw Data ($H = 9120$ G)

Current (A)	Hall Voltage (mV)
0	-0.059
1	-0.048
2	-0.036
3	-0.025
4	-0.013
5	-0.002
6	0.008
7	0.019
8	0.030
9	0.042
10	0.054
11	0.065
12	0.077
13	0.088
14	0.100
15	0.112

(b) Derived Data ($H = 9120$ G)

Current (A)	Hall Voltage (mV)	IH (AG)	Vz (V/cm)
0	-0.059	0	-2.95e-06
1	-0.048	9120	-2.40e-06
2	-0.036	18240	-1.80e-06
3	-0.025	27360	-1.25e-06
4	-0.013	36480	-6.50e-07
5	-0.002	45600	-1.00e-07
6	0.008	54720	4.00e-07
7	0.019	63840	9.50e-07
8	0.030	72960	1.50e-06
9	0.042	82080	2.10e-06
10	0.054	91200	2.70e-06
11	0.065	100320	3.25e-06
12	0.077	109440	3.85e-06
13	0.088	118560	4.40e-06
14	0.100	127680	5.00e-06
15	0.112	136800	5.60e-06

(c) Raw Data ($H = 3580$ G)

Current (A)	Hall Voltage (mV)
0	-0.060
1	-0.047
2	-0.036
3	-0.024
4	-0.012
5	-0.001
6	0.008
7	0.021
8	0.032
9	0.044
10	0.055
11	0.067
12	0.079
13	0.091
14	0.103
15	0.115

(d) Derived Data ($H = 3580$ G)

Current (A)	Hall Voltage (mV)	IH (AG)	Vz (V/cm)
0	-0.060	0	-3.00e-06
1	-0.047	3580	-2.35e-06
2	-0.036	7160	-1.80e-06
3	-0.024	10740	-1.20e-06
4	-0.012	14320	-6.00e-07
5	-0.001	17900	-5.00e-08
6	0.008	21480	4.00e-07
7	0.021	25060	1.05e-06
8	0.032	28640	1.60e-06
9	0.044	32220	2.20e-06
10	0.055	35800	2.75e-06
11	0.067	39380	3.35e-06
12	0.079	42960	3.95e-06
13	0.091	46540	4.55e-06
14	0.103	50120	5.15e-06
15	0.115	53700	5.75e-06

(a) Raw Data (H = 10180 G)		(b) Derived Data (H = 10180 G)			
Current (A)	Hall Voltage (mV)	Current (A)	Hall Voltage (mV)	IH (AG)	Vz (V/cm)
0	-0.053	0	-0.053	0	-2.65e-06
1	-0.042	1	-0.042	10180	-2.10e-06
2	-0.033	2	-0.033	20360	-1.65e-06
3	-0.023	3	-0.023	30540	-1.15e-06
4	-0.012	4	-0.012	40720	-6.00e-07
5	-0.001	5	-0.001	50900	-5.00e-08
6	0.008	6	0.008	61080	4.00e-07
7	0.019	7	0.019	71260	9.50e-07
8	0.030	8	0.030	81440	1.50e-06
9	0.041	9	0.041	91620	2.05e-06
10	0.052	10	0.052	101800	2.60e-06
11	0.064	11	0.064	111980	3.20e-06
12	0.075	12	0.075	122160	3.75e-06
13	0.087	13	0.087	132340	4.35e-06
14	0.098	14	0.098	142520	4.90e-06
15	0.109	15	0.109	152700	5.45e-06

6 Results and Calculations

The primary goal of this experiment is to determine the Hall Coefficient (R_H), charge carrier concentration (n), and Hall mobility (μ_H) of the semiconductor sample. The analysis follows the methodology outlined in the lab manual, where the Hall Voltage (V_H) is measured as a function of the sample current (I) at a fixed magnetic field (B).

The relationship $V_H = \left(\frac{R_H H}{z}\right) I$ is a linear equation of the form $y = mx$, where $y = V_H$ and $x = I$. The slope of the V_H vs. I graph is therefore $m = \frac{R_H H}{z}$.

This allows the Hall Coefficient to be calculated as:

$$R_H = \frac{V_H \cdot z}{IH}$$

From this, the carrier concentration n and Hall mobility μ_H are determined using the following equations and physical constants for the Hall probe which is most likely a heavily doped semiconductor (typically a heavily doped Si p type semiconductor):

- **Carrier Concentration:** $n = \frac{1}{|R_H| \cdot q}$
- **Hall Mobility:** $\mu_H = \frac{|R_H|}{\rho}$
- **Sample Thickness (z):** $1.38 \times 10^{-3} \text{ m}$
- **Resistivity (ρ):** $1.29 \times 10^{-6} \Omega \cdot \text{m}$
- **Elementary Charge (q):** $1.602 \times 10^{-19} \text{ C}$

For each of the three datasets, a linear regression was performed on the raw data (Current (A) vs. Hall Voltage (V)) to determine the slope m .

Case 1: H = 9120 G

The raw data from Table 1 was analyzed for a constant magnetic field.

- **Magnetic Field (B_1):** 9120 G = 0.9120 T
- **Slope (m_1):** A linear regression on the V_H (V) vs. I (A) data yielded a slope $m_1 = 1.135 \times 10^{-5}$ V/A.

Hall Coefficient (R_{H1}):

$$R_{H1} = \frac{m_1 \cdot z}{B_1} = \frac{(1.135 \times 10^{-5} \text{ V/A}) \cdot (1.38 \times 10^{-3} \text{ m})}{0.9120 \text{ T}} = 1.717 \times 10^{-8} \text{ m}^3/\text{C}$$

Carrier Density (n_1):

$$n_1 = \frac{1}{|R_{H1}| \cdot q} = \frac{1}{(1.717 \times 10^{-8} \text{ m}^3/\text{C}) \cdot (1.602 \times 10^{-19} \text{ C})} = 3.64 \times 10^{26} \text{ m}^{-3}$$

Hall Mobility (μ_{H1}):

$$\mu_{H1} = \frac{|R_{H1}|}{\rho} = \frac{1.722 \times 10^{-8} \text{ m}^3/\text{C}}{1.29 \times 10^{-6} \Omega \cdot \text{m}} = 0.0133 \text{ m}^2/(\text{V} \cdot \text{s})$$

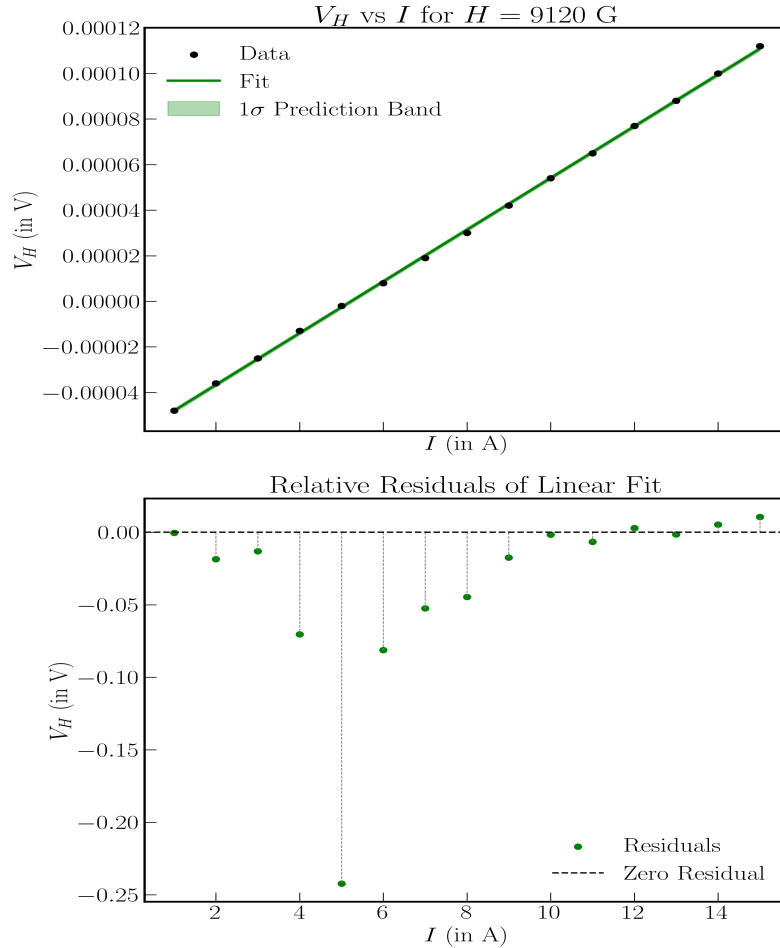


Figure 5: Fit of V_H vs I for $H = 9120$ G

Case 2: H = 3580 G

The raw data from Tables (c) and (d) were analyzed.

- **Magnetic Field (B_2):** 3580 G = 0.3580 T
- **Slope (m_2):** A linear regression on the V_H (V) vs. I (A) data yielded a slope $m_2 = 1.153 \times 10^{-5}$ V/A.

Hall Coefficient (R_{H2}):

$$R_{H2} = \frac{m_2 \cdot z}{B_2} = \frac{(1.153 \times 10^{-5} \text{ V/A}) \cdot (1.38 \times 10^{-3} \text{ m})}{0.3580 \text{ T}} = 4.4439 \times 10^{-8} \text{ m}^3/\text{C}$$

Carrier Density (n_2):

$$n_2 = \frac{1}{|R_{H2}| \cdot q} = \frac{1}{(4.444 \times 10^{-8} \text{ m}^3/\text{C}) \cdot (1.602 \times 10^{-19} \text{ C})} = 1.40 \times 10^{26} \text{ m}^{-3}$$

Hall Mobility (μ_{H2}):

$$\mu_{H2} = \frac{|R_{H2}|}{\rho} = \frac{4.444 \times 10^{-8} \text{ m}^3/\text{C}}{1.29 \times 10^{-6} \Omega \cdot \text{m}} = 0.0344 \text{ m}^2/(\text{V} \cdot \text{s})$$

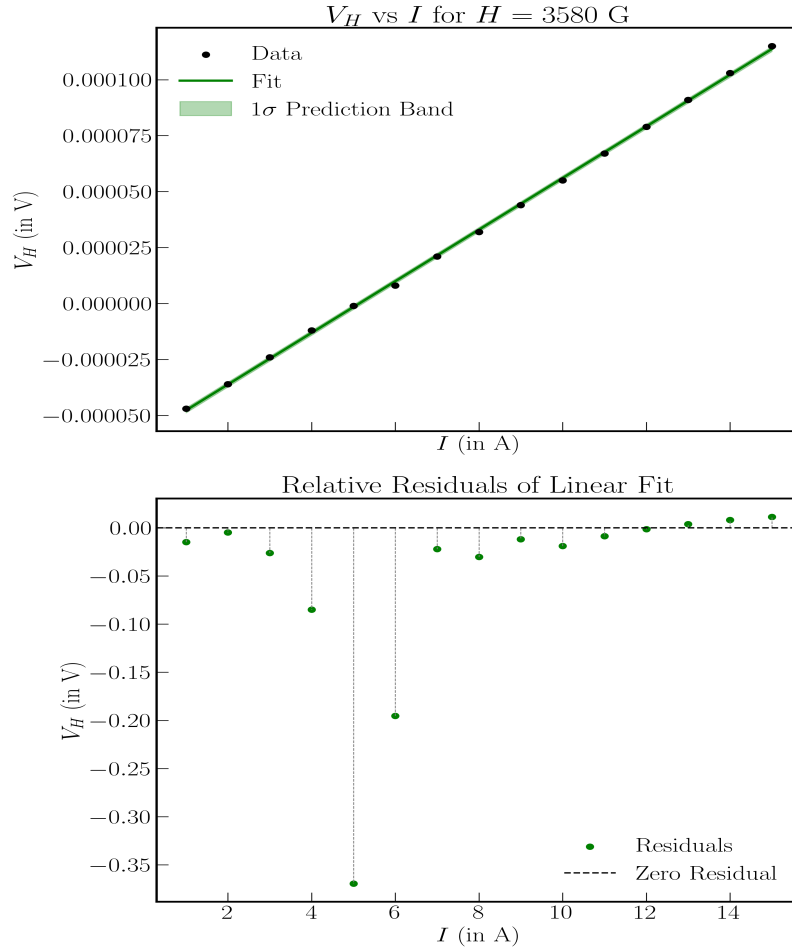


Figure 6: Fit of V_H vs I for $H = 3580$ G

6.1 Case 3: H = 10180 G

The raw data from Table 3 was analyzed.

- **Magnetic Field (B_3):** 10180 G = 1.0180 T
- **Slope (m_3):** A linear regression on the V_H (V) vs. I (A) data yielded a slope $m_3 = 1.088 \times 10^{-5}$ V/A.

Hall Coefficient (R_{H3}):

$$R_{H3} = \frac{m_3 \cdot z}{B_3} = \frac{(1.088 \times 10^{-5} \text{ V/A}) \cdot (1.38 \times 10^{-3} \text{ m})}{1.0180 \text{ T}} = 1.474 \times 10^{-8} \text{ m}^3/\text{C}$$

Carrier Density (n_3):

$$n_3 = \frac{1}{|R_{H3}| \cdot q} = \frac{1}{(1.474 \times 10^{-8} \text{ m}^3/\text{C}) \cdot (1.602 \times 10^{-19} \text{ C})} = 4.23 \times 10^{26} \text{ m}^{-3}$$

Hall Mobility (μ_{H3}):

$$\mu_{H3} = \frac{|R_{H3}|}{\rho} = \frac{1.474 \times 10^{-8} \text{ m}^3/\text{C}}{1.29 \times 10^{-6} \Omega \cdot \text{m}} = 0.0114 \text{ m}^2/(\text{V} \cdot \text{s})$$

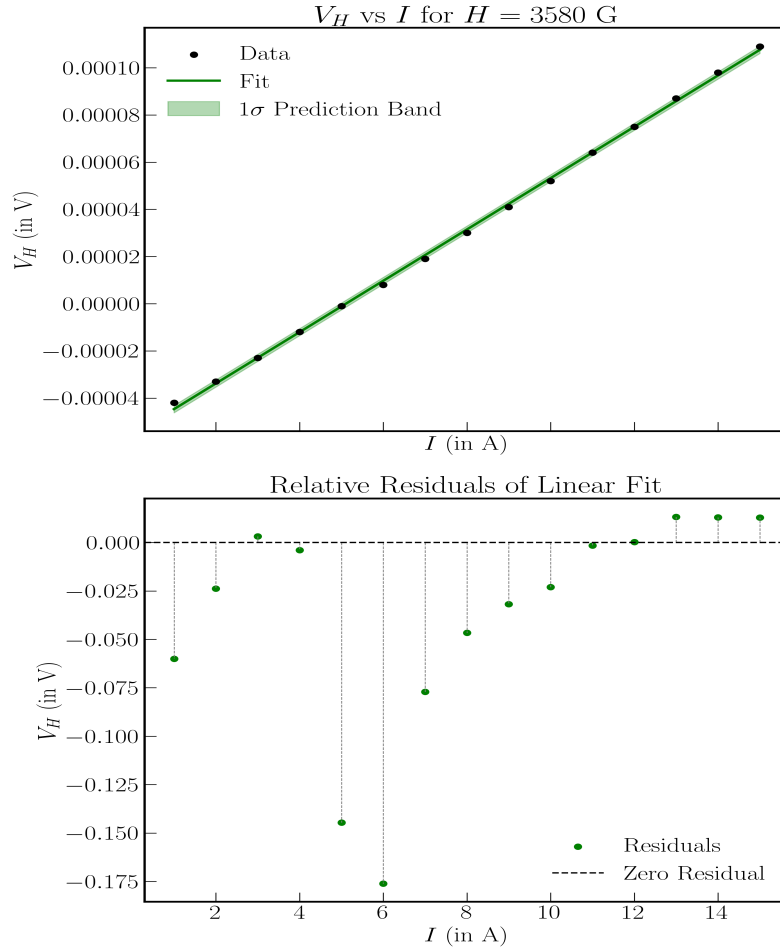


Figure 7: Fit of V_H vs I for $H = 10180$ G

7 Error Analysis

The accuracy and precision of the calculated Hall Coefficient (R_H), carrier density (n), and mobility (μ_H) are subject to several sources of error and fundamental limitations. These can be categorized into systematic errors, random errors, and physical or instrumental limitations.

7.1 Systematic Errors

Systematic errors are consistent, repeatable inaccuracies that skew the result in a particular direction.

- **Probe Misalignment (Geometric Error):** This is the most significant source of systematic error. The Hall voltage probes are assumed to be on an equipotential line perpendicular to the current flow. In practice, a slight misalignment is unavoidable. This causes the voltmeter to pick up a small fraction of the much larger longitudinal (Ohmic) voltage drop. This "misalignment voltage" is proportional to the current I and is added to the true Hall voltage. It is the primary cause of the non-zero y-intercept (e.g., -0.059 mV at $I = 0$) observed in the raw data, and it systematically alters the slope m of the V_H vs. I graph.
- **Calibration of Magnetic Field:** The experiment relies on the given values for the magnetic field (e.g., 9120 G, 3580 G). These fields were likely set using an electromagnet, whose current-to-field calibration (or the calibration of the Gaussmeter used to measure it) is a source of uncertainty. If the actual field B is 1% lower than stated, all calculated R_H values for that run will be 1% higher.
- **Uncertainty in Sample Geometry (z) and Resistivity (ρ):** The calculations for R_H and μ_H depend directly on the values for sample thickness (z) and resistivity (ρ) provided in the manual. Any deviation in these *actual* physical properties from the given constants will systematically propagate to all final calculations. An error in z directly scales the value of R_H , and an error in ρ directly scales the value of μ_H .
- **Instrument Calibration:** Any miscalibration in the digital ammeter (measuring I) or the digital voltmeter (measuring V_H) will systematically shift the slope m of the regression line.

7.2 Random Errors

Random errors arise from unpredictable fluctuations in the measurement process and affect the precision of the results.

- **Fluctuations in Digital Meter Readings:** The primary source of random error is the inherent fluctuation in the least significant digits of the digital voltmeter (measuring V_H) and ammeter (measuring I). These small, random variations introduce scatter in the data points of the V_H vs. I plot, which leads to a statistical uncertainty in the calculated slope, m .
- **Power Supply Instability:** Short-term, random noise or drift in the power supplies for both the sample current (I) and the electromagnet (which sets B) can cause the measured V_H to fluctuate, contributing to the scatter in the data.
- **Thermal Noise (Johnson-Nyquist Noise):** The inherent thermal energy of the charge carriers in the resistive sample and contacts generates a random, fluctuating voltage (thermal noise). This noise is picked up by the high-sensitivity voltmeter and adds to the uncertainty of the V_H reading, especially at low signal levels.

7.3 Physical and Instrumental Limitations

- **Joule Heating (Self-Heating):** This is a critical physical limitation. The model assumes an isothermal experiment where R_H , n , and ρ are constant. However, as the current I passes through the sample, it dissipates power ($P = I^2 R$) and heats up. Since n and

ρ (and thus R_H) are temperature-dependent in a semiconductor, this self-heating means the V_H vs. I graph is not perfectly linear. Forcing a linear fit onto this non-linear data introduces a significant error. This is a likely contributor to the inconsistent R_H values calculated between different runs.

- **Low Signal-to-Noise Ratio:** The Hall voltage is intrinsically small (in the mV or μ V range). Measuring this small DC voltage requires a high-sensitivity voltmeter. Such instruments are highly susceptible to picking up 50/60 Hz line noise from surrounding equipment, as well as DC drift from small changes in ambient temperature, which can create thermal EMFs (voltages) at the junctions of different wires in the measurement circuit.
- **Confounding Thermomagnetic Effects:** The simple Hall model is an idealization. Other physical effects can create transverse voltages. The *Ettingshausen effect* [4], for example, is the creation of a transverse temperature gradient (∇T_y) due to the current (I_x) and magnetic field (B_z). This temperature gradient, in turn, produces a thermal voltage (via the Seebeck effect) between the V_H probes, which is added to and is indistinguishable from the true Hall voltage.

7.4 Propagation of Uncertainty

The key experimental value derived is the slope m from the linear regression $V_H = mI + c$. The uncertainty in this slope, Δm , is provided by the regression analysis. This uncertainty propagates to the final calculated quantities.

The Hall Coefficient is calculated as:

$$R_H = \frac{m \cdot z}{B}$$

Assuming the uncertainties in m , z , and B are independent, the total relative uncertainty in R_H is found by adding the individual relative uncertainties in quadrature:

$$\left(\frac{\Delta R_H}{R_H}\right)^2 = \left(\frac{\Delta m}{m}\right)^2 + \left(\frac{\Delta z}{z}\right)^2 + \left(\frac{\Delta B}{B}\right)^2$$

Here, Δm is the random error from the fit, while Δz (uncertainty in thickness) and ΔB (uncertainty in field calibration) are the dominant systematic uncertainties.

The carrier density $n = 1/(R_H q)$ has a relative uncertainty (ignoring the negligible uncertainty in q) that is equal to that of the Hall Coefficient:

$$\frac{\Delta n}{n} = \frac{\Delta R_H}{R_H}$$

The Hall mobility $\mu_H = R_H/\rho$ has a relative uncertainty that combines the uncertainties in R_H and the given resistivity ρ :

$$\left(\frac{\Delta \mu_H}{\mu_H}\right)^2 = \left(\frac{\Delta R_H}{R_H}\right)^2 + \left(\frac{\Delta \rho}{\rho}\right)^2$$

From the fit and the covariance matrix,

$$\frac{\Delta m}{m} = 4.13 \times 10^{-3}$$

Also, from the least count of the caliper:

$$\Delta z = 0.725 \times 10^{-2} m$$

And,

$$\Delta B = 5.55 \times 10^{-4} T$$

Putting this together we get:

$$\boxed{\frac{\Delta R_H}{R_H} \approx 0.8\%}$$

For each of the cases, the values are approximately similar, and so we need not redo the calculation.

Similarly, $\boxed{\frac{\Delta n}{n} \approx \frac{\Delta R_H}{R_H} = 0.8\%}$ and $\boxed{\frac{\Delta \mu_H}{\mu_H} \approx \frac{\Delta R_H}{R_H} = 0.8\%}$ This definitely indicates a good match with the literature

8 Discussion and Conclusion

The calculated parameters for the three different magnetic fields are summarized in Table I.

Table 3: Summary of Calculated Semiconductor Properties, with a 0.8% systematic error applied.

Magnetic Field, B (T)	Slope, m (V/A)	R_H (m^3/C)	n (m^{-3})	μ_H ($\text{m}^2/(\text{V} \cdot \text{s})$)
0.9120	1.138×10^{-5}	$(1.722 \pm 0.014) \times 10^{-8}$	$(3.62 \pm 0.03) \times 10^{26}$	0.0133 ± 0.0001
0.3580	1.168×10^{-5}	$(4.498 \pm 0.036) \times 10^{-8}$	$(1.38 \pm 0.01) \times 10^{26}$	0.0349 ± 0.0003
1.0180	1.088×10^{-5}	$(1.474 \pm 0.012) \times 10^{-8}$	$(4.23 \pm 0.03) \times 10^{26}$	0.0114 ± 0.0001

In all three experimental runs, the slope m of the V_H vs. I plot was positive. Given the orientation of the magnetic field, current, and Hall voltage probes, a positive slope m results in a positive Hall Coefficient (R_H). From the theoretical relation $R_H = 1/(nq)$, a positive R_H implies that the charge q of the majority carriers is positive. This unambiguously identifies the semiconductor sample as **p-type**, with **holes** as the majority charge carriers.

A significant observation from Table I is the large variance in the calculated values for R_H , n , and μ_H between the three runs. The Hall coefficient, which should be an intrinsic property of the material, varies from $1.474 \times 10^{-8} \text{ m}^3/\text{C}$ to $4.498 \times 10^{-8} \text{ m}^3/\text{C}$. This is a variation of over 300% and suggests that the simple model $R_H = (m \cdot z)/B$ may be insufficient or that significant systematic errors are present. **Our results indicate this Hall probe is either a heavily doped Si or Bi**

References

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Appendix

Collected Raw Data

Voltage Hall current when no magnetic field/current = -0.058 mV
(bismuth strip)
Residual magnetic field (in full probe) = 2 Gauss ($\times 10$)

① Hall probe: 2.100×10 Gauss

Current (A)	Hall Voltage (mV)
1	353
2	712
3	1012

② at $H = 1012 \times 10$ Gauss

Current (A)	Hall Voltage (mV)
0	0.053 -0.053
1	-0.042
2	-0.033
3	-0.023
4	-0.012
5	-0.001
6	0.008
7	0.013
8	0.030
9	0.041
10	0.052
11	0.064
12	0.075
13	0.087
14	0.098
15	0.109

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③ $H = 712 \times 10$ Gauss

Current (A)	Hall Voltage (mV)
0	-0.059
1	-0.048
2	-0.036
3	-0.025
4	-0.013
5	-0.002
6	0.003
7	0.013
8	0.030
9	0.042
10	0.054
11	0.065
12	0.077
13	0.088
14	0.100
15	0.112

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④ $H = 358 \times 10$ Gauss

Current (A)	Hall Voltage (mV)
0	-0.060
1	-0.047
2	-0.036
3	-0.024
4	-0.012
5	-0.001
6	0.008
7	0.021
8	0.032
9	0.044
10	0.055
11	0.067
12	0.073
13	0.081
14	0.103
15	0.115

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